

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

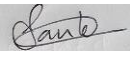
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I, SANDEEP K / 192521253 (Name / Reg No), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Students Signature: 

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:

Assessment Tasks Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical application. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear causeeffect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Analytical Problem Solving

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA1513

Course Outcome: CO2 – Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Question 1: Compare Type-1 and Type-2 Hypervisors for a Mission-Critical Cloud Datacenter

Introduction

A hypervisor is the virtualization layer that allows multiple virtual machines (VMs) to run on shared physical hardware. Hypervisors are broadly classified into Type-1 (bare-metal) and Type-2 (hosted), and the choice between them directly affects performance, security, and manageability in a datacenter hosting mission-critical applications such as banking or healthcare systems.

Type-1 Hypervisor (Bare-Metal)

- Runs directly on the physical hardware without an underlying host operating system (e.g., VMware ESXi, Microsoft Hyper-V, Citrix XenServer, KVM).
- Performance: Near-native performance because the hypervisor has direct access to CPU, memory, and I/O resources, with minimal overhead.
- Security: Smaller attack surface since there is no general-purpose host OS to exploit; strong isolation between VMs makes it suitable for multi-tenant environments.
- Manageability: Centralized management consoles (e.g., vCenter) support large-scale automation, live migration, and high availability, but require specialized administration skills.

Type-2 Hypervisor (Hosted)

- Runs as an application on top of a conventional host operating system (e.g., Oracle VirtualBox, VMware Workstation).
- Performance: Lower performance due to the additional abstraction layer of the host OS, causing extra CPU and I/O overhead.
- Security: Larger attack surface because vulnerabilities in the host OS can compromise all hosted VMs.
- Manageability: Easier to install and use for development or testing, but not designed for large-scale datacenter orchestration.

Justification and Recommendation

For a datacenter hosting mission-critical applications, a Type-1 hypervisor is the better choice. It delivers near-native performance required for high transaction volumes, provides stronger isolation and a reduced attack surface essential for regulatory compliance (e.g., HIPAA, PCI-DSS), and integrates with enterprise-grade management tools for high availability, load balancing, and disaster recovery. Type-2 hypervisors remain appropriate only for development, testing, or personal use where performance and security are secondary concerns.

Question 2: Workload Isolation for Finance, Healthcare, and Student Portals on Shared Hardware

How Virtualization Architecture Supports Isolation

- Hardware-level isolation: Each tenant's workload runs in a separate VM with its own virtual CPU, memory, and virtual NIC, enforced by the hypervisor so one VM cannot directly access another's resources.
- Hypervisor-enforced sandboxing: The hypervisor mediates all hardware access, preventing a compromised VM from breaking out (VM escape) into another tenant's VM.
- Network virtualization (VLANs/VXLAN, virtual switches, micro-segmentation): Logically separates traffic of finance, healthcare, and student-portal workloads even though they share the same physical network fabric.
- Storage virtualization: Uses separate virtual disks, storage pools, or encrypted logical volumes per tenant so data at rest remains isolated.
- Resource pools and quotas: CPU/memory reservations and limits prevent one workload (e.g., the student portal) from starving finance or healthcare VMs of resources (the 'noisy neighbor' problem).
- Role-based access control (RBAC) and identity management: Ensures administrators and applications can only manage the VMs and data belonging to their own tenant.

Two Major Attack Surfaces

- Hypervisor/VM-escape vulnerabilities: If an attacker compromises the hypervisor itself or exploits a VM-escape flaw, they could gain access to all co-located VMs, including highly sensitive finance and healthcare data making the hypervisor a single point of compromise for the entire shared infrastructure.
- Shared network and management plane: The virtual network fabric and centralized management/orchestration console (e.g., a compromised admin credential or an unsecured API) can be exploited to pivot laterally between tenants, intercept traffic, or reconfigure isolation boundaries such as VLAN tags or firewall rules.

Question 3: VM Host Utilization Snapshot – Bottleneck Analysis and Corrective Actions

Given Data

CPU utilization: 92% | Memory utilization: 88% | Storage I/O latency: 35 ms | Average VM boot time: 170 seconds

Analysis of Likely Bottlenecks

- CPU (92%): Sustained utilization above ~85% indicates CPU contention. VMs are likely experiencing CPU ready time (waiting in the scheduler queue), which slows down application response and increases boot time.
- Memory (88%): High memory usage close to the host's capacity risks memory ballooning or swapping to disk, which severely degrades performance and further increases I/O load.
- Storage I/O latency (35 ms): Typical acceptable latency for VM storage is under 10-20 ms; 35 ms suggests the storage backend (SAN/NAS or local disks) is overloaded or underprovisioned, causing slow read/write operations.
- VM boot time (170 s): An abnormally long boot time is a symptom, not a root cause it reflects the combined effect of CPU contention, memory pressure, and slow storage I/O all competing for the same host resources during the boot sequence.

Corrective Actions

- Use vMotion / live migration or DRS (Distributed Resource Scheduler) to rebalance VMs across additional hosts and reduce CPU contention.
- Right-size VMs by reclaiming over-allocated vCPUs and memory, and enable memory reservation for critical VMs to avoid ballooning/swapping.
- Upgrade or reconfigure storage: move to faster SSD/NVMe tiers, enable storage caching, or distribute I/O across additional datastores to bring latency below 20 ms.
- Implement resource pools with CPU/memory limits and reservations to guarantee capacity for critical workloads.
- Add monitoring and alerting thresholds (e.g., CPU >80%, latency >20 ms) for proactive capacity planning before performance degrades further.

Question 4: Software Defined Datacenter (SDDC) vs Traditional Datacenter Agility

Overview

A Software Defined Datacenter (SDDC) abstracts, pools, and automates all datacenter resources compute, storage, and network through software, delivering them as services. This is fundamentally more agile than a traditional datacenter, where resources are tightly coupled to physical hardware and provisioned manually.

Compute Virtualization

In a traditional datacenter, deploying a new server requires physical procurement and manual OS installation, taking days or weeks. In an SDDC, compute is virtualized so new VMs or containers can be provisioned in minutes through self-service portals or APIs, and workloads can be dynamically migrated (vMotion) or auto-scaled based on demand greatly improving agility and resource utilization.

Storage Virtualization

Traditional storage is siloed per application with fixed capacity, making expansion slow and costly. Software-Defined Storage (SDS) pools storage across the datacenter into a single logical resource that can be dynamically allocated, tiered, or expanded on demand, and supports

policybased automation (e.g., automatic replication or snapshotting) without manual reconfiguration of physical arrays.

Network Virtualization

Traditional networks depend on manual VLAN and switch/router configuration, which is slow to change and error-prone. Software-Defined Networking (SDN) and network virtualization (e.g., VMware NSX) decouple network services (routing, firewalling, load balancing) from physical hardware, allowing entire network topologies and micro-segmentation policies to be created or modified programmatically in minutes, enabling rapid, secure application deployment.

Conclusion

By virtualizing and automating compute, storage, and networking together, an SDDC converts a rigid, hardware-bound infrastructure into a flexible, software-controlled pool of resources enabling faster provisioning, elastic scaling, and consistent policy enforcement that a traditional datacenter cannot match.

Question 5: Multi-Cloud Federation Failure due to Identity Mismatches

Analysis of the Issue

In a multi-cloud federation, multiple independent cloud providers must trust each other's identity assertions so users can access resources across provider boundaries with a single identity. When identity mismatches occur such as differing user attribute schemas, inconsistent authentication protocols, or lack of a common trust anchor federated authentication fails, leading to access errors, duplicate identities, security gaps, or unauthorized access if trust is misconfigured.

Root Causes

- Absence of a common identity standard (e.g., one provider using SAML while another expects OAuth2/OIDC tokens).
- Inconsistent attribute mapping (e.g., role or department names differing between providers' directories).
- No centralized trust broker, causing each provider to validate identities independently and inconsistently.
- Lack of synchronized identity lifecycle management, so a deprovisioned user in one cloud may still have residual access in another.

Proposed Secure Identity and Privacy Design (Federation-Based)

- Adopt a standard federation protocol such as SAML 2.0 or OpenID Connect (OIDC)/OAuth 2.0, with a single Identity Provider (IdP) acting as the trusted source of authentication for all federated cloud providers (Service Providers).
- Implement Single Sign-On (SSO) so users authenticate once with the central IdP and receive a signed token/assertion that all clouds trust, eliminating protocol mismatches.
- Establish a common attribute schema and use claims-based identity so roles and permissions map consistently across all provider domains.
- Use a trust broker / identity federation hub to manage certificates, token signing keys, and trust relationships centrally, ensuring consistent validation policies.
- Apply the principle of least privilege and attribute-based access control (ABAC) so each cloud only receives the minimum identity attributes needed, preserving user privacy.

- Enforce synchronized identity lifecycle management (provisioning/de-provisioning) via standards like SCIM (System for Cross-domain Identity Management) so access is revoked consistently across all federated clouds.
- Encrypt identity tokens/assertions in transit (TLS) and at rest, and use short-lived tokens with refresh mechanisms to reduce the risk of credential misuse.

Conclusion

By standardizing on a single federation protocol, centralizing trust through an identity broker, and enforcing least-privilege, encrypted identity exchange, the multi-cloud federation can resolve identity mismatches and provide secure, privacy-preserving, single sign-on access across all participating cloud providers.